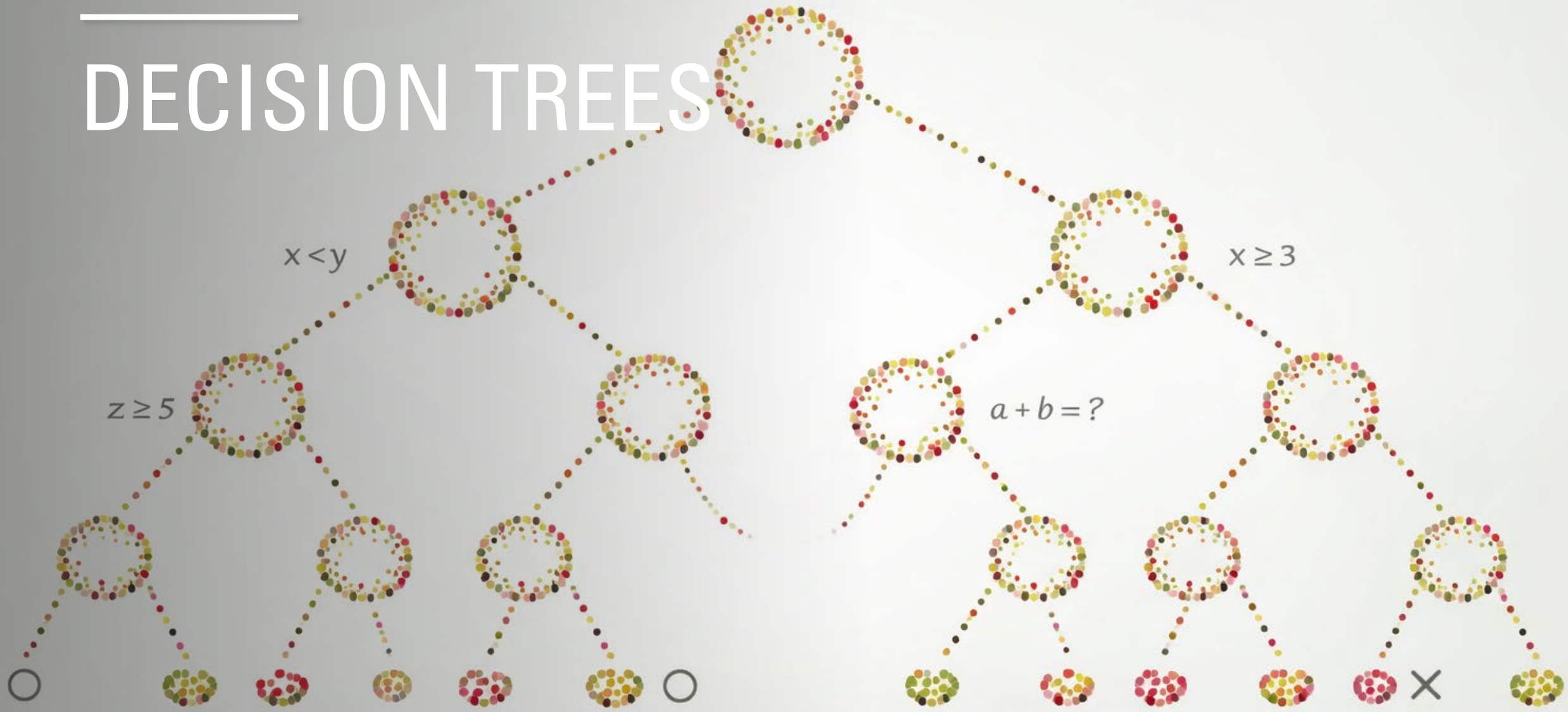
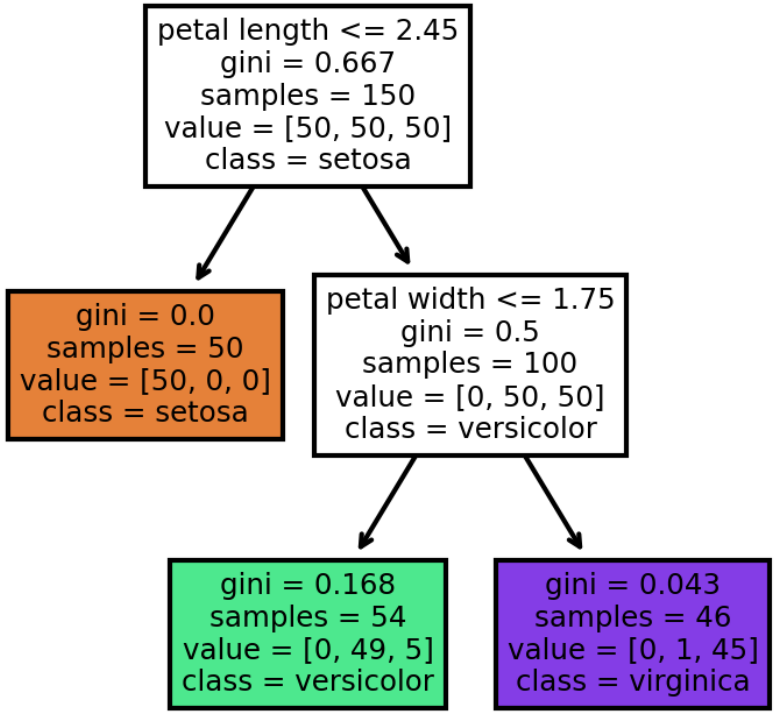
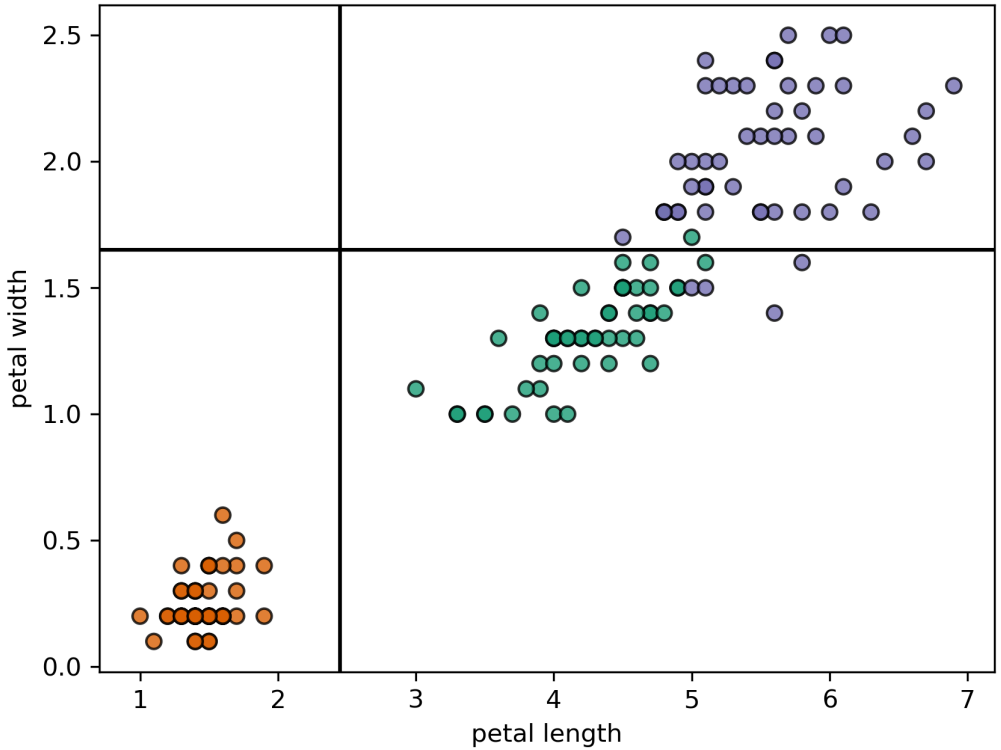
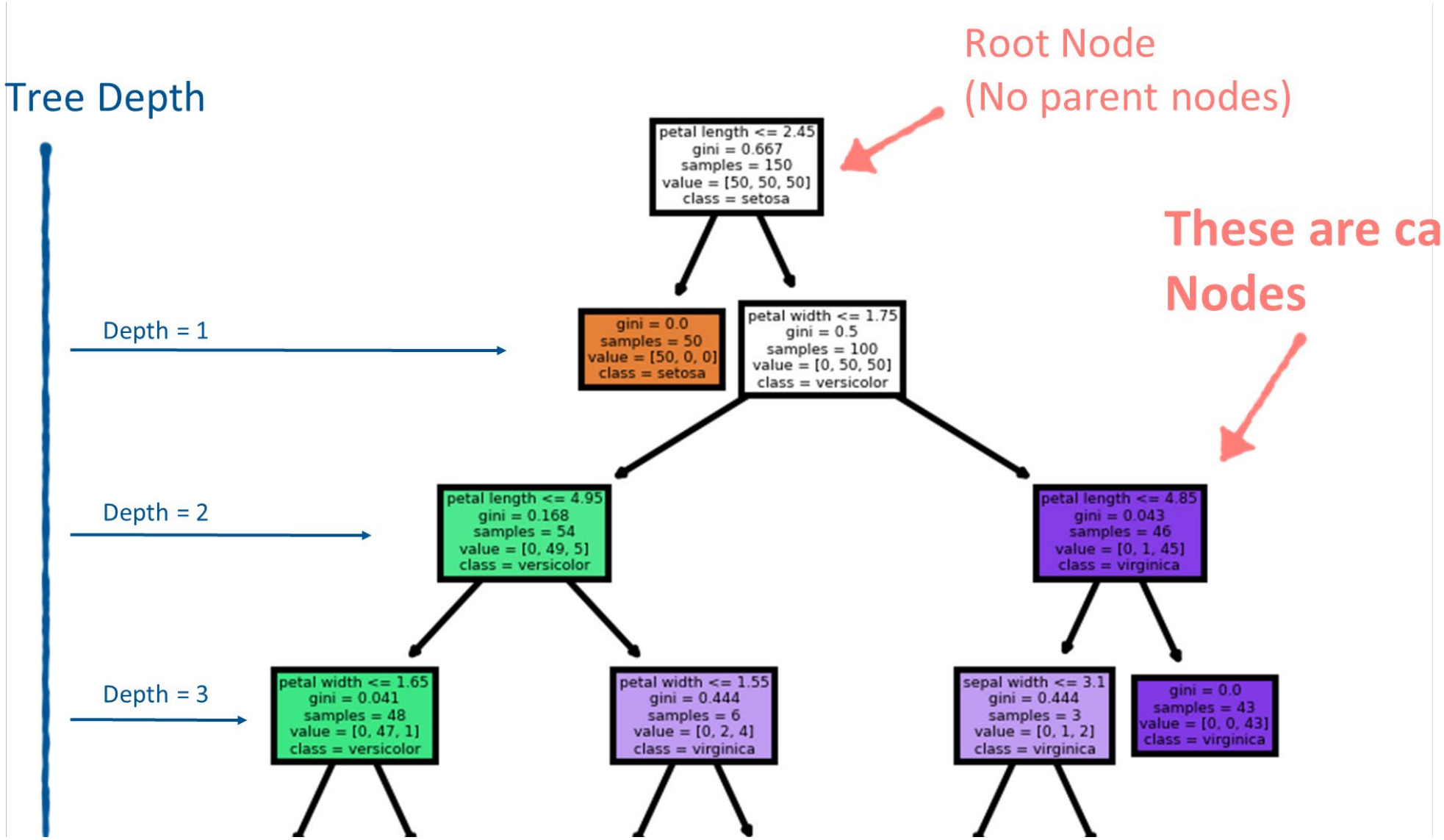
DECISION TREES



EXAMPLE DECISION TREE

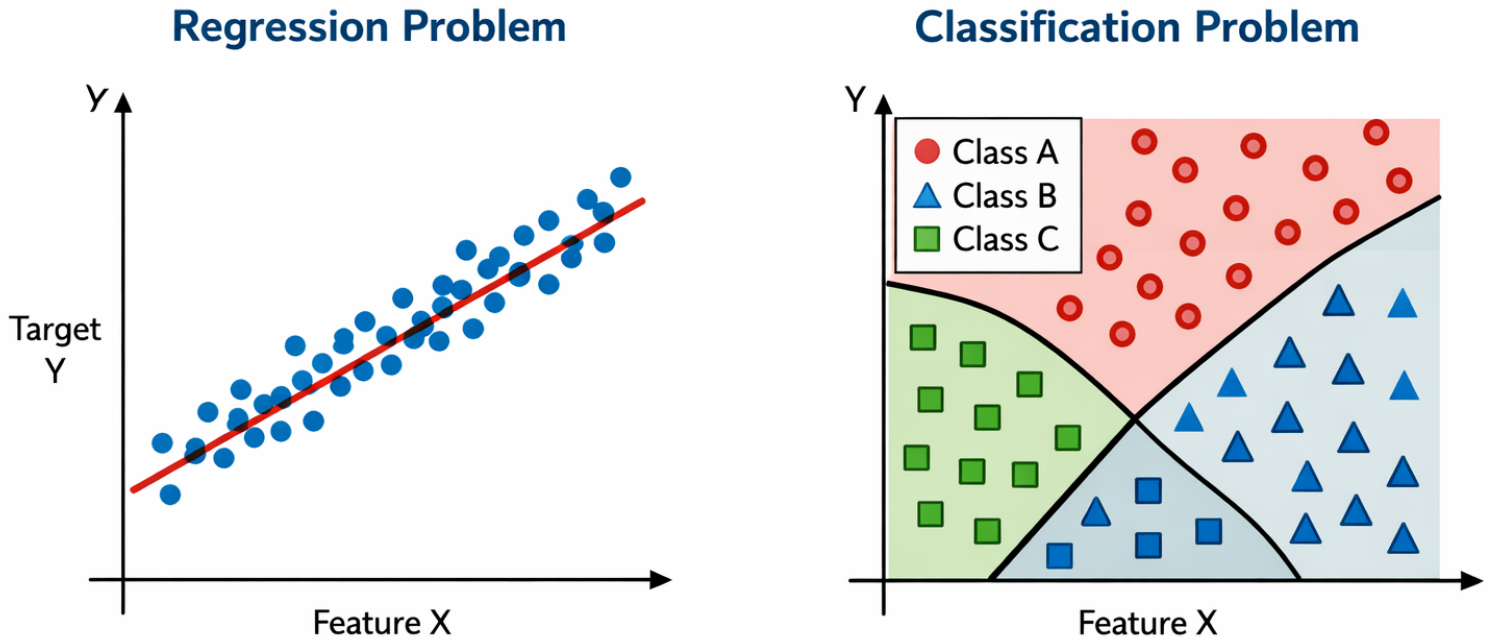





Tree Depth
Root Node
(No parent nodes)
These are called
Nodes
Depth = 1
Depth = 2
Depth = 3
petal length <= 2.45
gini = 0.667
samples = 150
value = [50, 50, 50]
class = setosa
gini = 0.0
samples = 50
value = [50, 0, 0]
class = setosa
petal width <= 1.75
gini = 0.5
samples = 100
value = [0, 50, 50]
class = versicolor
petal length <= 4.95
gini = 0.168
samples = 54
value = [0, 49, 5]
class = versicolor
petal length <= 4.85
gini = 0.043
samples = 46
value = [0, 1, 45]
class = virginica
petal width <= 1.65
gini = 0.041
samples = 48
value = [0, 47, 1]
class = versicolor
petal width <= 1.55
gini = 0.444
samples = 6
value = [0, 2, 4]
class = virginica
sepal width <= 3.1
gini = 0.444
samples = 3
value = [0, 1, 2]
class = virginica
gini = 0.0
samples = 43
value = [0, 0, 43]
class = virginica

HOW TO FIT?

- Binary decision search algorithm.
- Partition (split) data to optimize some objective.
- What might that objective be?



NODE IMPURITY MEASURES (CLASSIFICATION)

- Define p_{ik} as the proportion of instances in the ith node that are from class k
- Gini index

$$G_i = 1 - \sum_{k=1}^{K} p_{ik}^2 = \sum_{k=1}^{K} p_{ik}(1 - p_{ik})$$

- Cross Entropy

$$H_i = -\sum_{\substack{k=1 \\ if\ p_{ik} \neq 0}}^{K} p_{ik} \log_2(p_{ik})$$

NODE IMPURITY MEASURES (REGRESSION)

- Define $\overline{y}_i$ as the mean response of the ith node, which contains N_i instances
- Mean squared error (variance)

$$\text{MSE}_i = \frac{1}{N_i} \sum_{j \in R_i} \left(y_j - \overline{y}_i \right)^2$$

- Mean absolute error

$$\text{MAE}_i = \frac{1}{N_i} \sum_{j \in R_i} \left| y_j - \text{median}_i \right|$$

GROWING THE TREE (FITTING)

- Classification and Regression Tree (CART) algorithm:
  - Split the training data into two subsets using a single feature, k, and a threshold, t_k
  - Search for the (k, t_k) pair that minimizes the cost function:

CLASSIFICATION $\frac{n_{left}}{n} G_{left} + \frac{n_{right}}{n} G_{right}$ Where G is a measure of node impurity

REGRESSION $\frac{n_{left}}{n} MSE_{left} + \frac{n_{right}}{n} MSE_{right}$

GINI INDEX

- Range: 0 to $1 - \frac{1}{k}$ (k = number of classes)
- Zero impurity: All observations belong to one class.
- Maximum impurity: Classes are equally distributed.
  For binary classification, $G = 0.5$ at a 50–50 split.
- Penalizes mixed nodes and favors splits that isolate dominant classes.

GINI IMPURITY CALCULATIONS

- Gini index : $G_i = 1 - \sum_{k=1}^{K} p_{ik}^2$
- For the "green" node

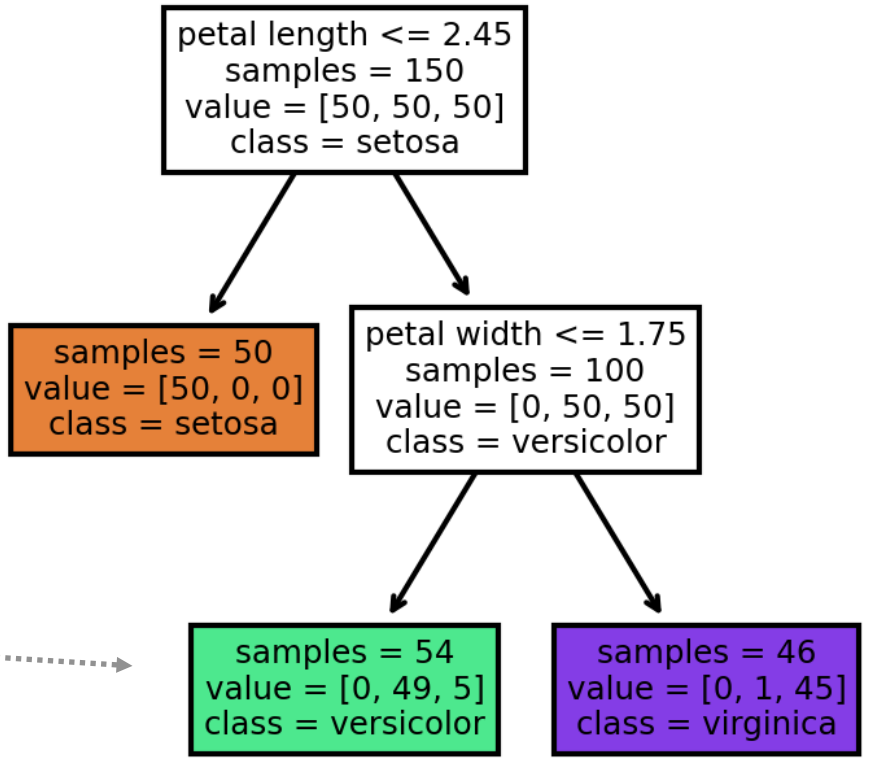
$$p_{i1} = (0/54);\ p_{i2} = (49/54);\ p_{i3} = (5/54)$$

$$G_i = 1 - ((0/54)^2 + (49/54)^2 + (5/54)^2)$$
$$= 0.168$$

- For the "purple" node

$$p_{i1} = (0/46);\ p_{i2} = (1/46);\ p_{i3} = (45/46)$$

$$G_i = 1 - ((0/46)^2 + (1/46)^2 + (45/46)^2)$$
$$= 0.043$$



ENTROPY

- Ranges between 0 and $\log_2(k)$ (where k is the number of classes)

  (for binary classification, when $k = 2$, max value is 1)
- Entropy is 0 when all elements belong to the same class, indicating perfect purity or “no disorder”
- It’s maximum value is reached when the classes are distributed as equally possible

  (for binary classification $H = 1$ indicates a perfect 50-50 split)

ENTROPY CALCULATIONS

- Entropy : $H_i = -\sum_{k=1}^{K} p_{ik} \log_2(p_{ik})$
- For the "green" node

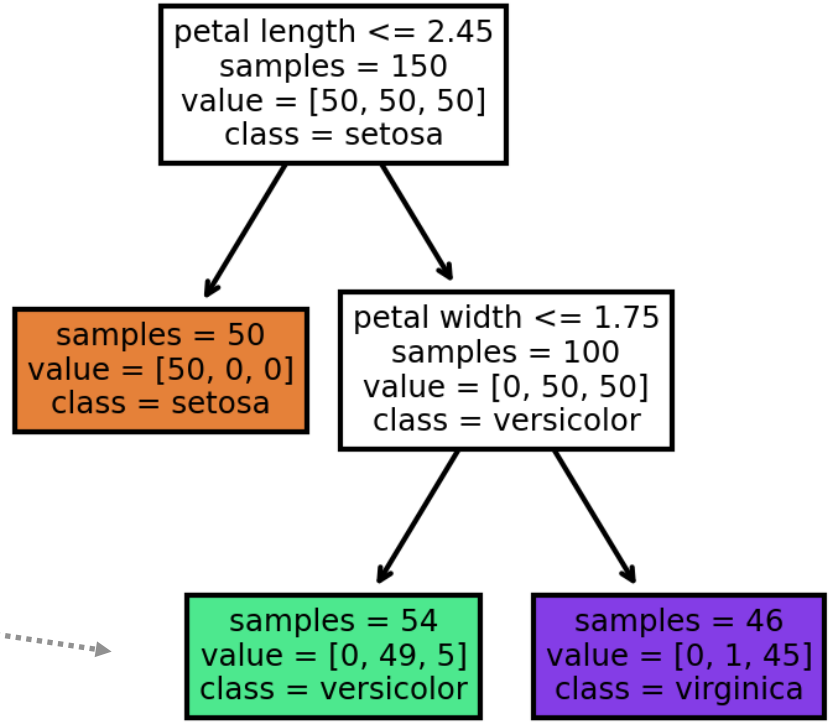
$$p_{i1} = (0/54);\ p_{i2} = (49/54);\ p_{i3} = (5/54)$$

$$H_i = -((49/54)\log_2(49/54) + (5/54)\log_2(5/54))$$
$$= 0.445$$

- For the "purple" node

$$p_{i1} = (0/46);\ p_{i2} = (1/46);\ p_{i3} = (45/46)$$

$$H_i = -((1/46)\log_2(1/46) + (45/46)\log_2(45/46))$$
$$= 0.151$$



GINI VS. ENTROPY

Node	Class counts	Gini index	Entropy (bits)
Green	[0, 49, 5]	0.168	0.445
Purple	[0, 1, 45]	0.043	0.151

- Both measures rank the purple node as purer. Their raw scores use different scales.
- Gini is faster to compute and is the scikit-learn default. Both usually produce similar trees.
- When they differ, **entropy** tends to produce more **balanced trees**, while **Gini** tends to isolate the most frequent class.

INFORMATION GAIN

- Entropy is used to calculate the **information gained** from the split
- Information gain:
  - Entropy(parent) – Entropy(split)

$$H(\text{parent}) = -\sum_{k=1}^{K} p_k \log p_k \qquad (1)$$

$$H_{\text{split}} = \sum_{j=1}^{J} \frac{n_j}{n} H(\text{child}_j) \qquad (2)$$

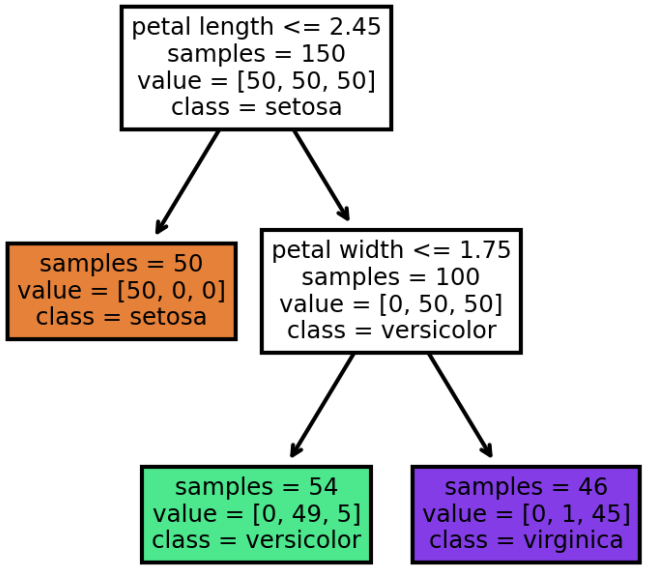
$$IG = H(\text{parent}) - H_{\text{split}} \qquad (3)$$

MAKING PREDICTIONS

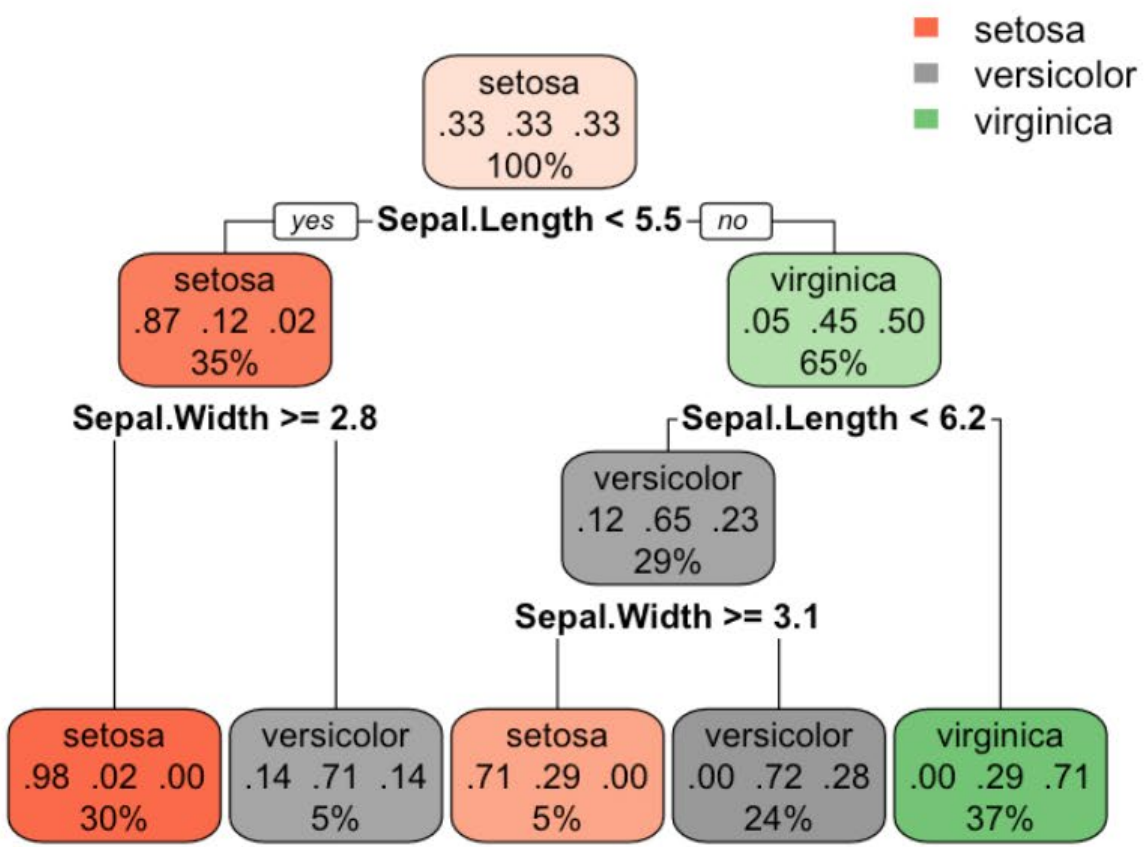
- We can predict the target of a new instance by following the tree from top to bottom
- At the terminal (leaf) nodes predict using:
  - Classification: The majority class in the node
  - Regression: The mean target value of the training values in the node

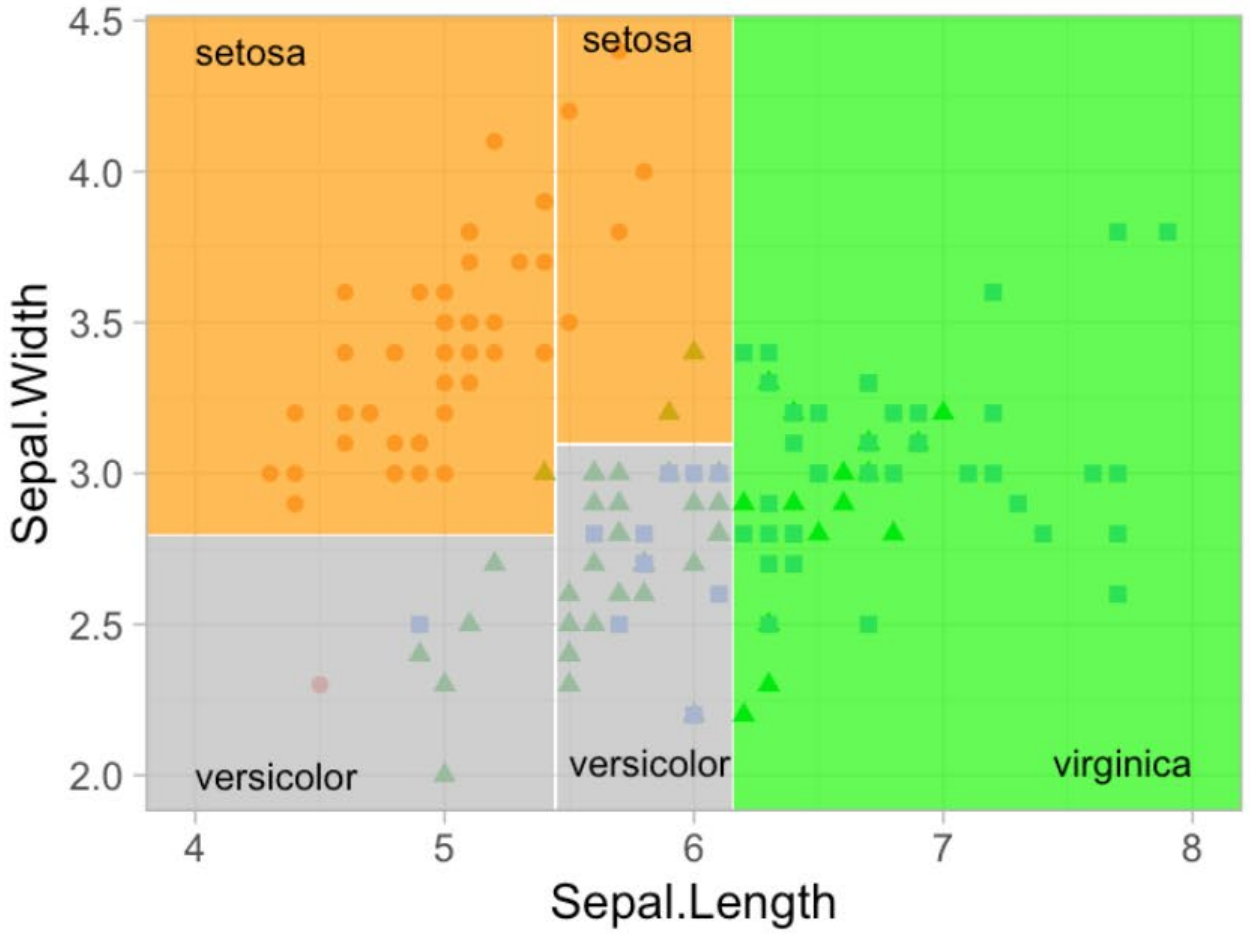
MAKING PREDICTIONS (CLASSIFICATION)

- We can predict the target of a new instance by following the tree from top to bottom
- At the terminal (leaf) nodes predict using:
  - Classification: The majority class in the node

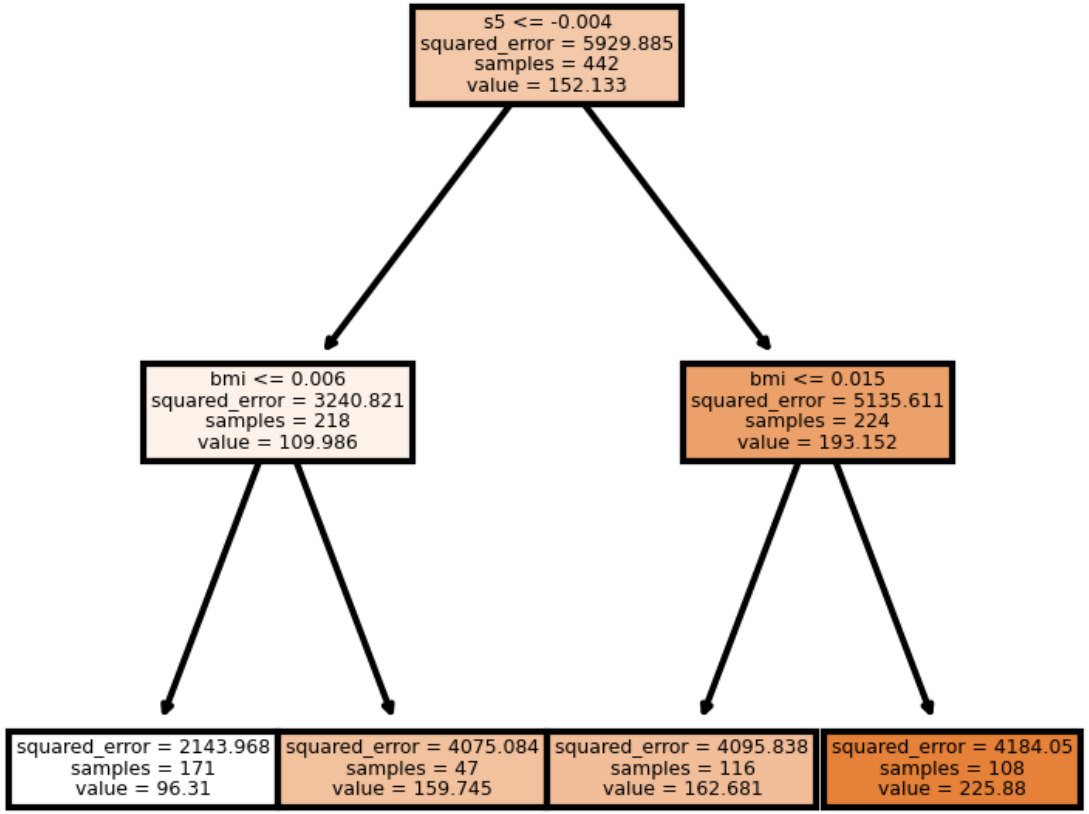


CLASSIFICATION BOUNDARY EXAMPLE

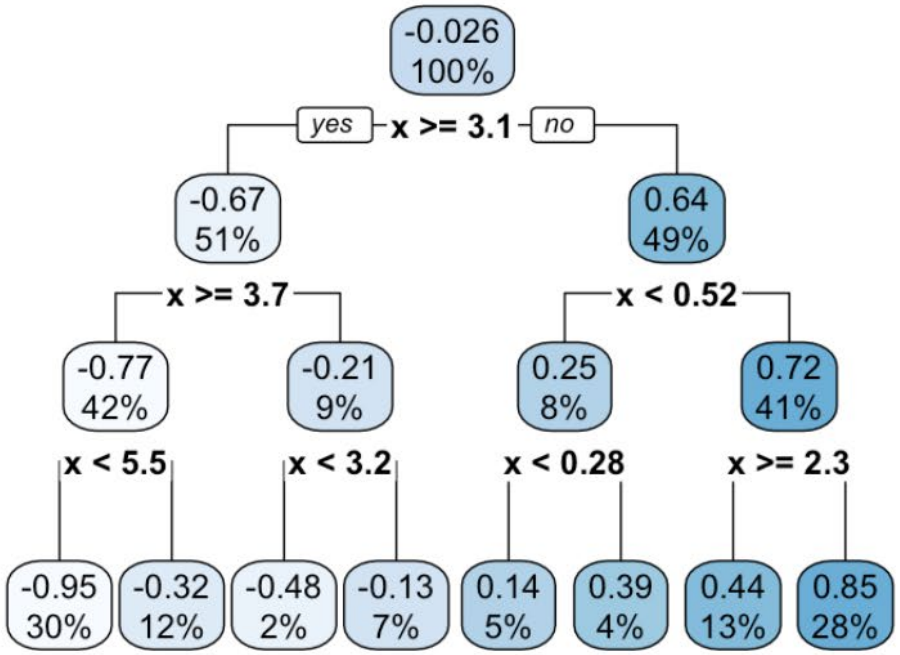


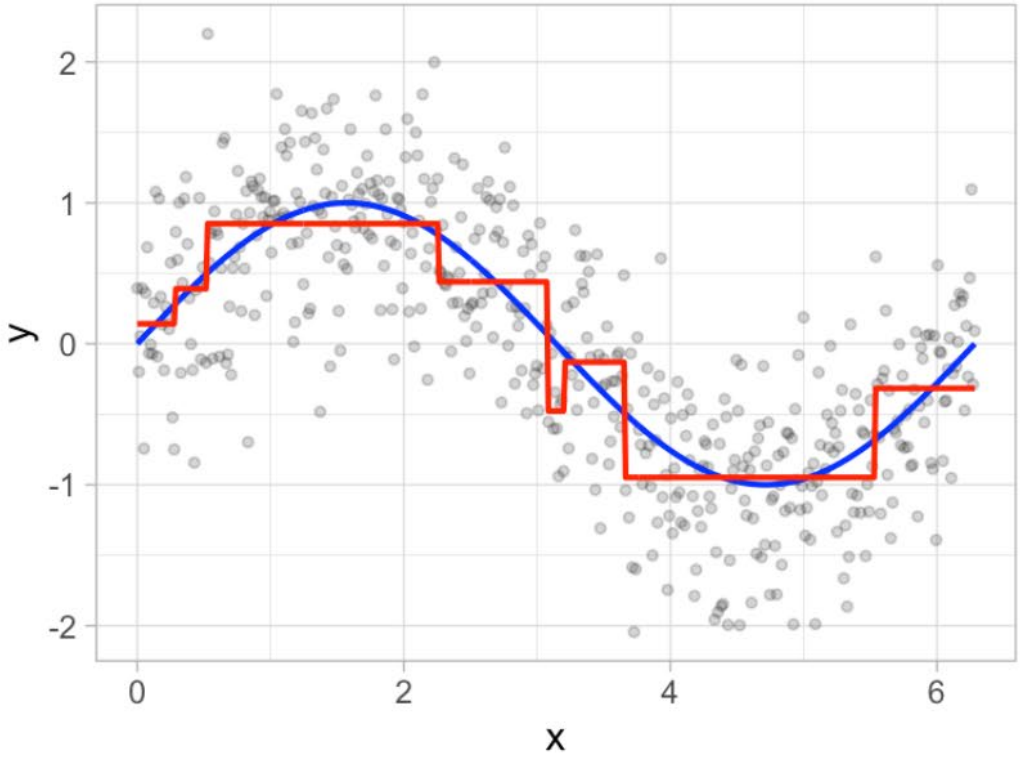


MAKING PREDICTIONS (REGRESSION)

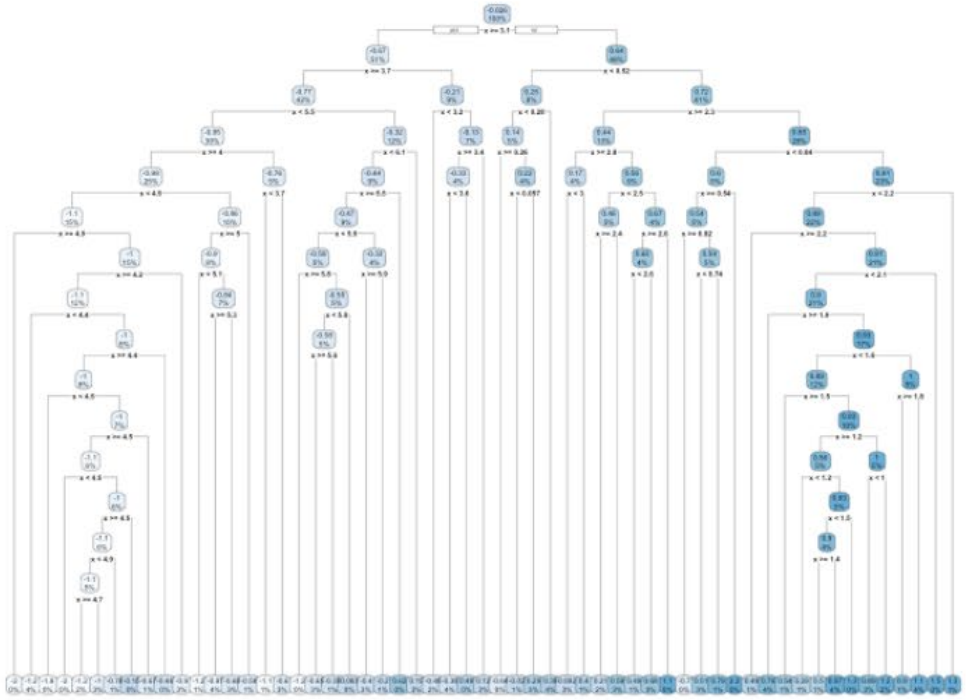


EXAMPLE: REGRESSION DECISIONS





EXAMPLE: REGRESSION DECISIONS



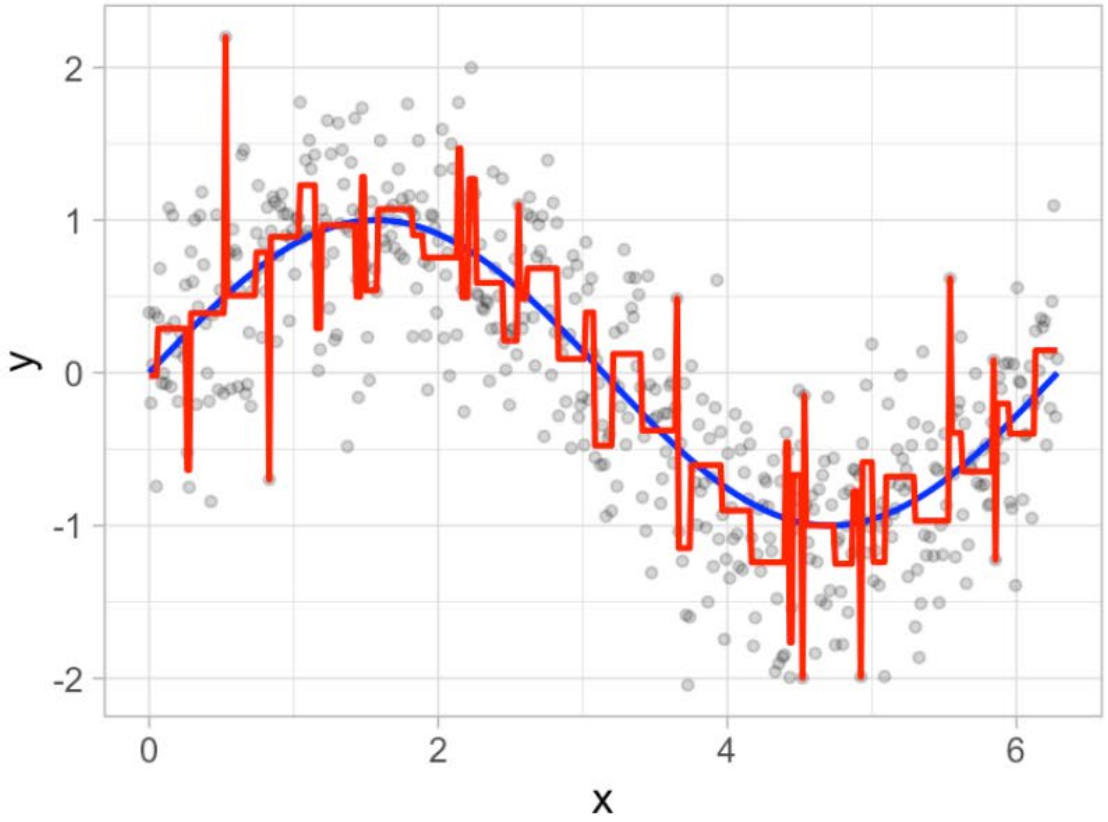
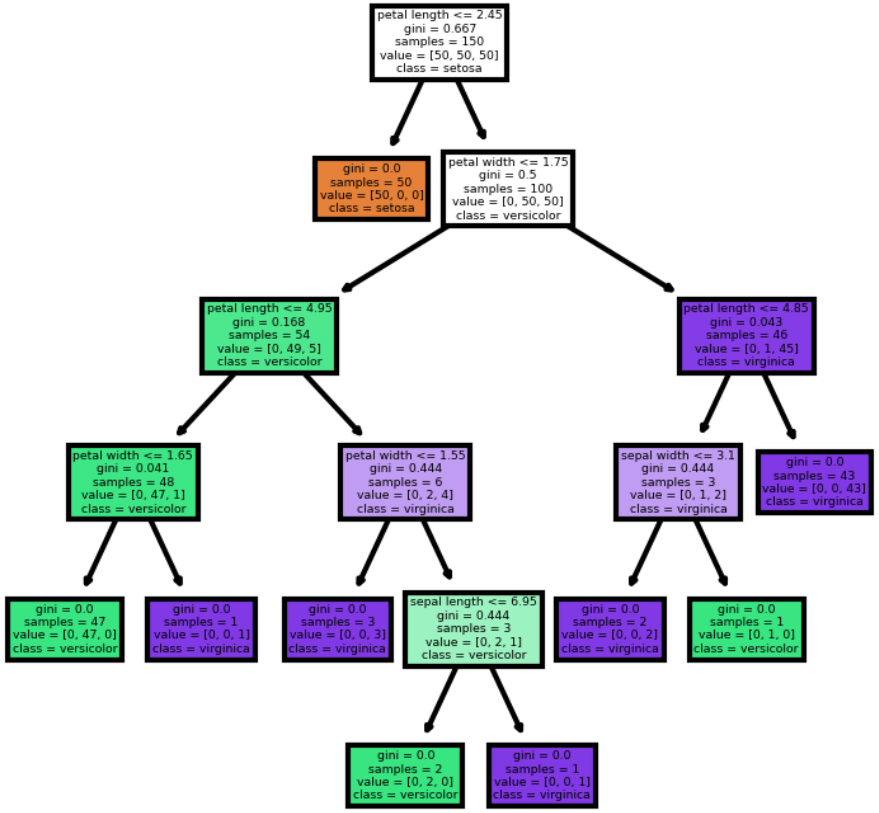


Figure 9.6: Overfit decision tree with 56 splits.

Trees can grow very large if left unconstrained



REGULARIZATION

- Recall that regularization means constraining the model to prevent overfitting
- Decision Trees can be regularized by
  - Constraining growth (pre-pruning)
  - Post-pruning the unconstrained tree (cost complexity pruning)

POST-PRUNING REGULARIZATION

- Cost complexity pruning (CCP)
  - Remove nodes that do not significantly reduce impurity
  - Prune nodes from bottom up if it results in a net improvement in the cost-complexity measure
    $R_\alpha(T) = R(T) + \alpha|\tilde{T}|$ where
    -$|\tilde{T}|$ is the number of terminal nodes and
    - $R(T)$ is the total impurity (Gini, Entropy, MSE)
    - $\alpha \geq 0$ is the regularization parameter controlling the trade-off
  - CCP finds the subtree that minimizes $R_\alpha(T)$
- **`ccp_alpha`**: subtree with largest $R_\alpha(T)$ that is smaller than `ccp_alpha` will be chosen

COST-COMPLEXITY PRUNING (CCP): INTUITION

Minimize $R_\alpha(T) = R(T) + \alpha|T|$

- Why does increasing α make trees smaller?
- α multiplies the number of leaves.
- Larger $\alpha \rightarrow$ larger penalty for each additional leaf.
- A split must reduce impurity **enough** to justify its complexity cost.
- If impurity reduction is small, the penalty dominates $\rightarrow$ subtree is pruned.

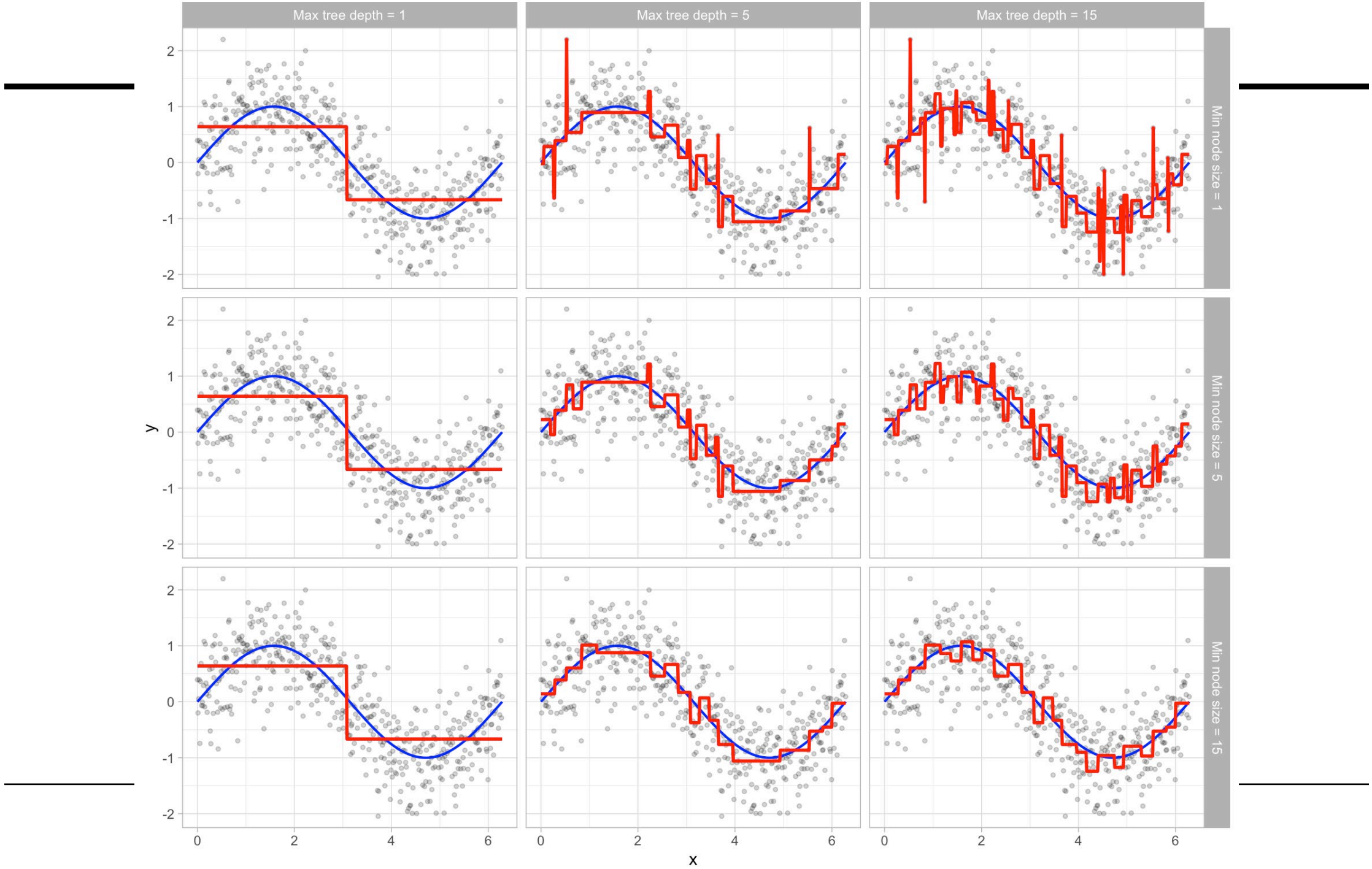
PRE- VERSUS POST-PRUNING

- Choice depends on:
  - Specific dataset and overfitting behavior observed
  - Domain knowledge
  - Model performance on a validation set
  - Personal preference

PRE-PRUNING REGULARIZATION

- Stop tree growth before it perfectly classifies the training data
  - **`max_depth`**: maximum depth of tree
  - **`min_samples_split`**: minimum number of samples required to split an internal node.
  - **`min_samples_leaf`**: minimum number of sample required to be at a leaf node
  - **`max_leaf_nodes`**: maximum number of leaf nodes
  - **`min_impurity_decrease`**: a node will be split if the split induces a decrease of impurity greater than or equal to this value

reduce "`max`" parameters and/or increase "`min`" parameters if tree is overfitting


Max tree depth = 1
Max tree depth = 5
Max tree depth = 15
Min node size = 1
Min node size = 5
Min node size = 15
y
x
2
1
0
-1
-2
0
2
4
6

OTHER DECISION TREE ALGORITHMS

- CART is a very popular tree algorithm but it isn't the only one
  (though it is the only one in scikit-learn)
- Other algorithms:
  - **ID3** - designed for classification problems
    - Uses information gain to select the feature that best splits the set of items
    - Works only with categorical features
    - More than two branches can be created from a node
  - **C4.5** - extension of ID3 that handles both continuous and discrete features
  - **C5.0** - improved version of C4.5
  - **CHAID** - uses chi-square tests to determine the best split

PROS AND CONS OF TREES

- Pros
  - Easy to build and explain
  - Non-parametric and can easily capture non-linear relationships
  - Very little data preprocessing is required

- Cons
  - Tendency to overfit
  - Difficulty with linear relationships
  - *High variance models (training set selection has a huge impact on the resulting tree!)*